KATALEYA

v2.4 — Development Milestone Reference

March 2026 | For Future-Self Reference | github.com/kwasikontor45/kataleya

PURPOSE
Complete state of the Kataleya codebase as of v2.4 — what was built, what changed, what broke and how it was fixed, and every key source file added during the redesign sprint. Written for the founding developer to pick up exactly where this left off.

— CURRENT STATE —

v2.4 — Where We Are

The architecture is complete. The UI has been fully redesigned with the NeonCard system and GhostPulseOrb. The app is running on a real device via Expo Go, served from Crostini Linux on a Chromebook. The codebase is version-controlled at github.com/kwasikontor45/kataleya.

Key identifiers:

```
GitHub:       github.com/kwasikontor45/kataleya
Expo account: bleedin6ed6e
EAS project:  72297d8e-1e5e-45f4-9e04-da70ac5d7207
Bundle ID:    com.kataleya.app  (iOS + Android)
Repo on device: ~/kataleya/artifacts/kataleya-app
Start command:  npx expo start --lan  (phone + Chromebook same WiFi)
```

— FEATURE STATUS —

Complete Feature Table

Feature	Status
Three-vault storage (Surface 11 keys / Sanctuary SQLite / Fortress 8 keys)	✅ **Complete**
Circadian engine (4 phases: dawn / day / goldenHour / night)	✅ **Complete**
GhostPulseOrb — animated central visual, replaces orchid SVG	✅ **v2.4**
NeonCard — foundation card component (cyan/violet/amber/pink)	✅ **v2.4**
Sobriety timer (second-precise) + 10 milestone markers	✅ **Complete**
Haptic bloom profiles (Budding → Parting → Labellum → Full Bloom)	✅ **Complete**
Mood logging (5 states, circadian phase, restlessness score)	✅ **Complete**
Private journal (4,000 chars, delete-on-demand, keyboard-safe)	✅ **Complete**
Growth screen — glow dot milestone timeline + confidence insight cards	✅ **v2.4**

Insights engine (3 pattern types, confidence-gated ≥ 72%)	✅ **Complete**
Vault screen — NeonCard tiers, backup flow, burn ritual	✅ **v2.4**
AES-256 encrypted backup (zero-knowledge, timingSafeEqual)	✅ **Needs EAS build**
Burn Ritual (3-phase linear, tombstone crash recovery)	✅ **Complete**
Bridge screen + tombstone recovery at launch	✅ **Complete**
Onboarding (7 steps, progress dots)	✅ **Complete**
Push notifications (daily reminder + milestones)	✅ **iOS / Android needs EAS**
Sponsor channel — invite / accept / poll / presence / check-in	✅ **v2.4 NeonCard**
Water + Light hold buttons with GIF background fade on press	✅ **v2.4**
E2E encrypted messages (X25519 + XSalsa20-Poly1305 via TweetNaCl)	✅ **Complete**
Responsive heart (BPM 45–78, system state, biometric sampling)	✅ **Complete**
DataBridge cycling words (sampling → encoding → secure…)	✅ **v2.4**
Day count pulse / breath animation (1.0 → 1.04 cycle)	✅ **v2.4**
Breathing exercise — staggered ring animation (ripple effect)	✅ **v2.4**
Butterfly card (surfaces on avg mood ≤ 4.5 over last 5 logs)	✅ **v2.4**
Predictive suggestion card (local, phase-aware, mood-gated)	✅ **v2.4**
Cover screen + Privacy screen + Error boundary	✅ **Complete**
Dark mode override + circadian auto-switch	✅ **Complete**
Subscription / payment layer (PAYWALL_ACTIVE=false)	🔜 **v3.0 — flip to activate**
EAS production build → App Store + Play Store	🔜 **Needs Apple ($99) + Google ($25)**
iCloud Drive backup automation	🔜 **v3.0**
RevenueCat integration	🔜 **v3.0**

— V2.4 CHANGES —

What Was Built This Sprint

New Components

NeonCard.tsx

Foundation card component replacing all plain View cards across every screen. Accepts accentRgb (RGB triplet string e.g. "127,201,201") or glowColor ("cyan" | "violet" | "amber" | "pink" | "neutral"). Controls borderIntensity and fillIntensity independently. Pressable variant wraps TouchableOpacity automatically. Exports NEON_RGB constant with the four phase-matched RGB triplets.

GhostPulseOrb.tsx

Replaces OrchidProgress as the central visual on the sanctuary home screen. Three concentric rings pulse at staggered rates (delays 0 / 600ms / 1200ms). Core ✧ glyph scales with BPM from useResponsiveHeart. Ring size grows with milestone stage: seed=62px core up to 112px at day 365. Color shifts with circadian phase. Outer ring ripples faster when restlessnessScore is high. Stage labels: waiting / beginning / awakening / reaching / opening / blooming / full bloom / radiant.

Redesigned Screens

- index.tsx — GhostPulseOrb replaces orchid. Day count number breathes with pulse animation. Phase-accent colors throughout. NeonCard mindfulness tiles (breathe/ground). Predictive suggestion card (local, phase-aware). Butterfly card on sustained low mood.
- journal.tsx — Neon mood pill selection state (each pill shows its mood color). Left accent bar on history entries encodes mood. Restlessness bar visible when score > 0.15. All cards NeonCard.
- growth.tsx — Glow dot milestone timeline (active dot 12px, achieved 8px, unachieved dim). Confidence-scaled NeonCard insight cards (fillIntensity = 0.04 + confidence × 0.1).
- sponsor.tsx — NeonCard hold buttons for water/light. GIF background fades in at 28% opacity on hold. Role picker, invite flow, message thread, garden path, disconnect card all NeonCard. Disconnect uses red NeonCard.
- vault.tsx — Fortress=amber, Sanctuary=cyan, Surface=violet NeonCards. Backup warning in amber NeonCard. Burn card in red NeonCard.

Animation Updates

- DataBridge cycling words: state label below ..: :.. cycles every 2.2s. present → encoding → sampling → reading → secure. Each system state has its own word set.
- BreathingExercise staggered rings: two outer rings animate 180ms and 360ms behind core — creates a ripple expanding outward instead of all pulsing in unison.
- Day count breathes: gentle 1.0 → 1.04 → 1.0 scale on a 4.4-second cycle using Easing.inOut(Easing.sin).

Media Assets

- assets/images/water.gif — Ocean waves at sunset. Fades in at 28% opacity behind Water hold button when pressed. Fades out on release. Stays 3s after signal fires then fades.
- assets/images/light.gif — Golden sun reflection on water. Same fade mechanic behind Light hold button.
- assets/images/butterfly.jpg — Butterfly sheltering under leaf in rain. Surfaces as a quiet NeonCard on the sanctuary home when last 5 mood logs average ≤ 4.5. Dark overlay (55% opacity) for text readability. "i know" dismiss button.

— HURDLES & RESOLUTIONS —

What Broke and How It Was Fixed

expo-sqlite plugin error on start	package.json had canary SDK 56 packages (expo-sqlite@56.0.0-canary-...) but expo itself was on SDK 54. The config plugin lookup failed because the canary build had a different internal structure.	Downgraded all canary packages to stable SDK 54 equivalents: expo-sqlite@13.0.0, expo-notifications@0.29.14, expo-secure-store@14.0.1, expo-sensors@14.0.2, expo-sharing@12.0.1, expo-document-picker@12.0.2, expo-file-system@18.0.12 Also removed "expo-sqlite" from plugins array in app.json — not needed for Expo Go dev, only for EAS native builds.
pnpm install EACCES symlink error	Repo was stored at ~/chaos-as-architecture/lab-one-projects/kataleya/ which maps to /mnt/chromeos/MyFiles/ — the ChromeOS FUSE filesystem. FUSE does not support symlinks and pnpm requires symlinks for its node_modules structure.	Copied the entire repo into the Crostini Linux filesystem: cp -r ~/chaos-as-architecture/lab-one-projects/kataleya ~/kataleya The Linux home directory (~/ inside Crostini) is a native ext4 filesystem that fully supports symlinks. All pnpm work happens here. Git pushes to GitHub keep both locations in sync.
sudo expo start broke file permissions	Running sudo npx expo start created cache and config files owned by root inside the project directory. Subsequent non-sudo runs could not write to those files.	Fixed ownership and cleared Metro cache: sudo chown -R bonesaw:bonesaw ~/kataleya sudo chown -R bonesaw:bonesaw ~/.expo sudo rm -rf /tmp/metro-* Rule going forward: never use sudo with expo or pnpm. Always run as regular user.
ngrok tunnel repeatedly timing out	ngrok was having an outage and/or Chromebook network configuration was blocking the tunnel handshake. The error "remote gone away" indicates the ngrok server dropped the connection.	Switched to --lan mode: npx expo start --lan Phone and Chromebook must be on the same WiFi network. LAN mode is faster, more reliable, and has no external dependency. No tunnel service involved.

npm install -g EACCES permission error	The system npm global prefix pointed to /usr/lib/node_modules which is root-owned. Installing global packages requires sudo, which then creates root-owned files.	Set a user-owned npm global prefix: mkdir -p ~/.npm-global npm config set prefix ~/.npm-global echo 'export PATH=~/.npm-global/bin:$PATH' >> ~/.bashrc source ~/.bashrc All subsequent npm install -g commands install to ~/.npm-global without sudo.
git push rejected — divergent branches	Changes were pushed from both Codespaces and Crostini environments simultaneously, creating divergent commit histories on the same branch.	git config pull.rebase false # set merge strategy once git pull origin main # creates a merge commit git push origin main # now succeeds This is a standard divergent branch resolution. The merge commit is harmless.
assets not found after git pull	The water.gif, light.gif, and butterfly.jpg files were pushed to GitHub from Codespaces. The ~/kataleya copy in Crostini was cloned before that push so it didn't have those files.	git pull origin main This fetched the new assets. The require() calls in sponsor.tsx and index.tsx could then resolve the image paths.
cp fails with (tabs) in path	The shell interprets parentheses as subshell syntax. Also, ~ does not expand when inside double quotes.	Two rules: 1. Escape parentheses: app/\(tabs\)/index.tsx 2. Never put ~ inside quotes — use it unquoted: cp source.tsx ~/kataleya/artifacts/kataleya-app/app/\(tabs\)/index.tsx
SQLite openDatabaseSync error in Expo Go	react-native-quick-crypto (used for backup AES-256) and the synchronous SQLite API both require JSI (JavaScript Interface) which Expo Go strips out. Only native EAS builds include JSI modules.	Expected limitation — not a bug. The error is non-fatal for UI development. All five tabs render correctly. Features gated on this: • AES-256 encrypted backup (needs EAS) • Android push notifications (needs EAS) Everything else works in Expo Go including all animations, sponsor

		channel, mood logging, journal, growth screen, and vault.

git stash pop restored conflicting package versions	Stashing before a pull saved the Codespaces dev-environment package versions (downgraded canary packages, expo-sqlite removed from plugins). After the pull, stash pop restored them, creating a modified state.	These are valid changes — they're what makes the app run in Codespaces/Crostini. Committed them explicitly: git add artifacts/kataleya-app/app.json git add artifacts/kataleya-app/package.jso n git add pnpm-lock.yaml git commit -m "fix: SDK 54 stable packages for dev environment"

— ENVIRONMENT —

Development Setup — Crostini

Working configuration

```
Device:         Chromebook + Crostini Linux (bonesaw@penguin)
Repo:           ~/kataleya/  (Crostini ext4 filesystem)
                Also at ~/chaos-as-architecture/lab-one-projects/kataleya/
GitHub:         github.com/kwasikontor45/kataleya
Expo account:   bleedin6ed6e
EAS project ID: 72297d8e-1e5e-45f4-9e04-da70ac5d7207
Bundle ID:      com.kataleya.app  (iOS + Android)
Android pkg:    com.kataleya.app
Node:           v20.x
pnpm:           v10.33.0
npm prefix:     ~/.npm-global  (user-owned, no sudo needed)
```

Start Expo

```
cd ~/kataleya/artifacts/kataleya-app
npx expo start --lan        # primary — phone + Chromebook same WiFi
npx expo start --tunnel     # if ngrok is available (often flaky)

# If permission errors after a sudo run:
sudo chown -R bonesaw:bonesaw ~/kataleya
sudo rm -rf /tmp/metro-*
npx expo start --lan
```

Git workflow

```
cd ~/kataleya
git add .
git commit -m "description"
git push origin main

# If rejected (divergent branches):
git config pull.rebase false
git pull origin main        # accept merge commit
git push origin main
```

Copy files with (tabs) in path

```
# Escape parentheses, never put ~ inside quotes
cp source.tsx ~/kataleya/artifacts/kataleya-app/app/\(tabs\)/index.tsx
cp component.tsx ~/kataleya/artifacts/kataleya-app/components/NeonCard.tsx
```

— NEXT STEPS —

v3.0 Roadmap

#	Item	Notes
1	**Apple Developer Account ($99/yr)**	Required for TestFlight and App Store. Prerequisite for all iOS native builds.
2	**Google Play Developer Account ($25)**	One-time fee. Required for Play Store distribution.
3	**EAS first build**	eas build --platform ios --profile development → TestFlight. Unlocks AES-256 backup and Android push notifications. Before building: add expo-sqlite back to app.json plugins.
4	**Restore expo-sqlite to app.json plugins**	Was removed to fix Expo Go startup. Must be added back before any EAS build.
5	**RevenueCat integration**	Flip PAYWALL_ACTIVE=true in utils/entitlements.ts. Wire RevenueCat SDK. Seed/Bloom/Garden tiers activate immediately.
6	**iCloud Drive backup automation**	Replace the manual share sheet with a silent iCloud Drive write on backup creation.
7	**Forward secrecy (double ratchet)**	Upgrade E2E messages from static X25519 to Signal-style double ratchet. Current is server-confidential which is sufficient for launch.

— SOURCE CODE —

New Files Added in v2.4

All files below are new to v2.4 — they did not exist before this sprint. Existing files modified (index.tsx, journal.tsx, growth.tsx, sponsor.tsx, vault.tsx, BreathingExercise.tsx, DataBridge.tsx) are in the GitHub commit history.

components/NeonCard.tsx

```
'use no memo';
import React from 'react';
import { View, TouchableOpacity, ViewStyle } from 'react-native';
import { ThemeTokens } from '@/constants/theme';

export type NeonGlowColor = 'cyan' | 'violet' | 'amber' | 'pink' | 'neutral';

export const NEON_RGB: Record<NeonGlowColor, string> = {
  cyan:    '127,201,201',  // day / water / breathing
  violet:  '155,109,255',  // night / grounding / journal
  amber:   '232,197,106',  // golden hour / light signal / Fortress vault
  pink:    '208,96,122',   // dawn
  neutral: '255,255,255',  // no tint
};

interface Props {
  children: React.ReactNode;
  theme: ThemeTokens;
  accentRgb?: string;        // e.g. '127,201,201'  (wins over glowColor)
  glowColor?: NeonGlowColor;
  onPress?: () => void;
  onLongPress?: () => void;
  style?: ViewStyle;
  borderIntensity?: number;  // 0-1, default 0.18
  fillIntensity?: number;    // 0-1, default 0.06
  disabled?: boolean;
}

export function NeonCard({
  children, theme, accentRgb, glowColor, onPress, onLongPress,
  style, borderIntensity = 0.18, fillIntensity = 0.06, disabled = false,
}: Props) {
  const rgb = accentRgb ?? (glowColor ? NEON_RGB[glowColor] : NEON_RGB.neutral);
  const cardStyle: ViewStyle = {
    backgroundColor: `rgba(${rgb}, ${fillIntensity})`,
    borderWidth: 1,
    borderColor: `rgba(${rgb}, ${borderIntensity})`,
    borderRadius: 16,
    overflow: 'hidden',
  };
  if (onPress || onLongPress) {
    return (
      <TouchableOpacity onPress={onPress} onLongPress={onLongPress}
        disabled={disabled} activeOpacity={0.75} style={[cardStyle, style]}>
        {children}
```

```
      </TouchableOpacity>
    );
  }
  return <View style={[cardStyle, style]}>{children}</View>;
}
```

components/GhostPulseOrb.tsx — key logic

```
// Core size scales with milestone stage
function orbSizeForDays(days: number): number {
  if (days >= 365) return 112;  // radiant
  if (days >= 180) return 104;  // full bloom
  if (days >= 90)  return 96;   // blooming
  if (days >= 30)  return 88;   // opening
  if (days >= 14)  return 80;   // reaching
  if (days >= 7)   return 74;   // awakening
  if (days >= 1)   return 68;   // beginning
  return 62;                    // waiting (seed)
}

// Props
interface Props {
  theme: ThemeTokens; phase: CircadianPhase; daysSober: number;
  restlessnessScore?: number;  // 0-1 — makes outer ring faster
  systemState?: string;        // 'holding' | 'present' | 'attuned' | 'celebrating'
  bpm?: number;                // drives core pulse speed
}

// Three staggered ring animations
const outerDuration = Math.max(1600, 3200 - restlessnessScore * 1200);
const coreDuration  = Math.round((60 / bpm) * 1000);
// ring1: outerDuration        delay: 0ms
// ring2: outerDuration * 1.25  delay: 600ms
// ring3: outerDuration * 1.6   delay: 1200ms
// core:  coreDuration          delay: 0ms

// Interpolated values
const ring1Opacity = ring1Anim.interpolate({ inputRange:[0,1],
outputRange:[0.12,0.28] });
const ring2Opacity = ring2Anim.interpolate({ inputRange:[0,1],
outputRange:[0.08,0.20] });
const ring3Opacity = ring3Anim.interpolate({ inputRange:[0,1],
outputRange:[0.05,0.14] });
const coreOpacity  = coreAnim.interpolate({ inputRange:[0,1],
outputRange:[0.65,1.0]  });
const coreScale    = coreAnim.interpolate({ inputRange:[0,1],
outputRange:[0.97,1.04] });

// Ring sizes relative to coreSize
const ring1Size = coreSize + 20;
const ring2Size = coreSize + 40;
const ring3Size = coreSize + 64;
```

DataBridge.tsx — cycling words

```
const STATE_CYCLES: Record<string, string[]> = {
  holding:     ['holding', 'sampling', 'listening', 'waiting', 'still'],
  present:     ['present', 'encoding', 'sampling', 'reading', 'secure'],
  attuned:     ['attuned', 'syncing', 'encoding', 'aligned', 'resonant'],
  celebrating: ['celebrating', 'transmitting', 'radiant', 'blooming', 'open'],
};

// Cycles every 2200ms, resets when systemState changes
useEffect(() => {
  const words = STATE_CYCLES[systemState] ?? [systemState];
  cycleIndex.current = 0;
  setCycleWord(words[0]);
  const interval = setInterval(() => {
    cycleIndex.current = (cycleIndex.current + 1) % words.length;
    setCycleWord(words[cycleIndex.current]);
  }, 2200);
  return () => clearInterval(interval);
}, [systemState]);
```

index.tsx — day count pulse animation

```
const dayPulse = useRef(new Animated.Value(1)).current;
useEffect(() => {
  const breathe = () => {
    Animated.sequence([
      Animated.timing(dayPulse, {
        toValue: 1.04, duration: 2200,
        easing: Easing.inOut(Easing.sin), useNativeDriver: true,
      }),
      Animated.timing(dayPulse, {
        toValue: 1.0, duration: 2200,
        easing: Easing.inOut(Easing.sin), useNativeDriver: true,
      }),
    ]).start(({ finished }) => { if (finished) breathe(); });
  };
  breathe();
  return () => dayPulse.stopAnimation();
}, [dayPulse]);

// In render:
<Animated.Text style={[styles.dayCount, {
  color: `rgba(${accentRgb}, 0.95)`,
  transform: [{ scale: dayPulse }]
}]}>
  {sobriety.daysSober}
</Animated.Text>
```

BreathingExercise.tsx — staggered ring animation

```
// Two additional ring anims — each breathes at different offset
const ring1Scale = useRef(new Animated.Value(1.0)).current;
```

```
const ring2Scale = useRef(new Animated.Value(1.0)).current;

// Inside animateToPhase() — fires before the core animation:
setTimeout(() => {
  Animated.timing(ring1Scale, {
    toValue: p.scale * 1.06, duration: 1100,
    easing: Easing.inOut(Easing.sin), useNativeDriver: true,
  }).start();
}, 180);  // 180ms behind core

setTimeout(() => {
  Animated.timing(ring2Scale, {
    toValue: p.scale * 1.12, duration: 1300,
    easing: Easing.inOut(Easing.sin), useNativeDriver: true,
  }).start();
}, 360);  // 360ms behind core

// Each ring uses its own scale:
// <Animated.View transform={[{ scale: scaleAnim }]} />   ← core orb
// <Animated.View transform={[{ scale: ring1Scale }]} />  ← inner ring
// <Animated.View transform={[{ scale: ring2Scale }]} />  ← outer ring
```

sponsor.tsx — GIF background on hold buttons

```
const WATER_IMG = require('@/assets/images/water.gif');
const LIGHT_IMG = require('@/assets/images/light.gif');

// HoldButton receives bgImage prop (any)
// Additional animated value inside HoldButton:
const imgOpacity = useRef(new Animated.Value(0)).current;

// On pressIn — fade in at 28% opacity:
Animated.timing(imgOpacity, {
  toValue: 0.28, duration: 600, useNativeDriver: true,
}).start();

// On pressOut / cancel — fade out:
Animated.timing(imgOpacity, {
  toValue: 0, duration: 400, useNativeDriver: true,
}).start();

// After fire — stays 3s then fades:
setTimeout(() => {
  Animated.timing(imgOpacity, {
    toValue: 0, duration: 800, useNativeDriver: true,
  }).start();
}, 3000);

// Rendered as absolute fill behind the progress fill bar:
{bgImage && (
  <Animated.Image source={bgImage}
    style={[styles.holdBgImage, { opacity: imgOpacity }]}
    resizeMode='cover' />
)}

// Style:
holdBgImage: {
  position: 'absolute', top: 0, left: 0, right: 0, bottom: 0,
```

```
  width: '100%', height: '100%',
},
```

index.tsx — butterfly card

```
const BUTTERFLY_IMG = require('@/assets/images/butterfly.jpg');

// Trigger check on mount
useEffect(() => {
  const check = async () => {
    try {
      const logs = await Sanctuary.getMoodLogs(10);
      if (logs.length < 4) return;
      const avg = logs.slice(0, 5)
        .reduce((s, l) => s + l.moodScore, 0) / 5;
      if (avg <= 4.5) setShowButterfly(true);
    } catch {}
  };
  check();
}, []);

// Render — only when showButterfly && !butterflyDismissed
<NeonCard theme={theme} glowColor='violet' borderIntensity={0.2}
fillIntensity={0.06}>
  <Image source={BUTTERFLY_IMG} style={styles.butterflyImage} resizeMode='cover' />
  <View style={styles.butterflyOverlay}>  {/* rgba(14,12,20,0.55) */}
    <Text style={styles.butterflyQuote}>
      butterflies rest when it rains{"\n"}because it damages their wings.
    </Text>
    <Text style={styles.butterflyQuoteSub}>
      it's okay to rest during the storms of life.{"\n"}
      you will fly again when it's over.
    </Text>
    <TouchableOpacity onPress={() => setButterflyDismissed(true)}>
      <Text style={styles.butterflyDismissText}>i know</Text>
    </TouchableOpacity>
  </View>
</NeonCard>

// butterflyImage:   width: '100%', height: 180
// butterflyOverlay: padding: 16, gap: 8, background rgba(14,12,20,0.55)
// Dismissed for session only — reappears next launch if mood still low
```